IX AUSTRALIA

INTERNET PEERING FOR AUSTRALIA

Network Flows for Peering Analysis

Introduction

- Member-owned, not-for-profit association
- Operator of 5 IXPs across Australia
- 400Gbit aggregate traffic peak
- 600+ peering ports (1G, 10G, 40G, 100G)
- 300+ member VLLs

Flows

- Total per-port usage is only so useful.
- Smarter capacity and connectivity planning requires deeper insight.
- sFlow, Netflow, IPFIX - all will accomplish the task given the right technology stack.

Flows for IX Australia

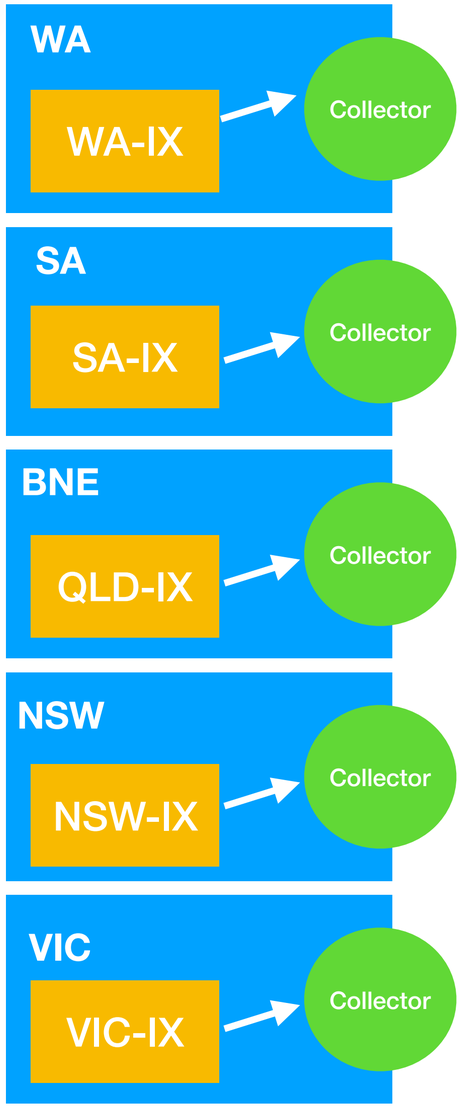
- Provide a service to members - assist them in making smart decisions about their networks.
- Multilateral Peering vs Bilateral Peering vs VLL traffic.
- Content consumption per-source.

Challenges

- Scale - 600+ ports and 400+ Gbit of traffic
- Transport - export, representation, distribution
- Enrichment - matching data to peers
- Storage - ingest, query, aggregation
- Analysis

Collect at Scale

- 45 devices producing sFlow data
  - ~0.5-1Mbit/sec of flow data per 10Gbit* (1:2000 sampling)
  - ~2-4Mbit/sec of flow data per 10Gbit* (1:512 sampling)
- Original n-tuple information in packet header
  (MPLS, VLAN-wrapped IPv4/v6 packet header).
- **Solution:**
  - Collect close to the source and only collect information that's useful for now



PMACCT

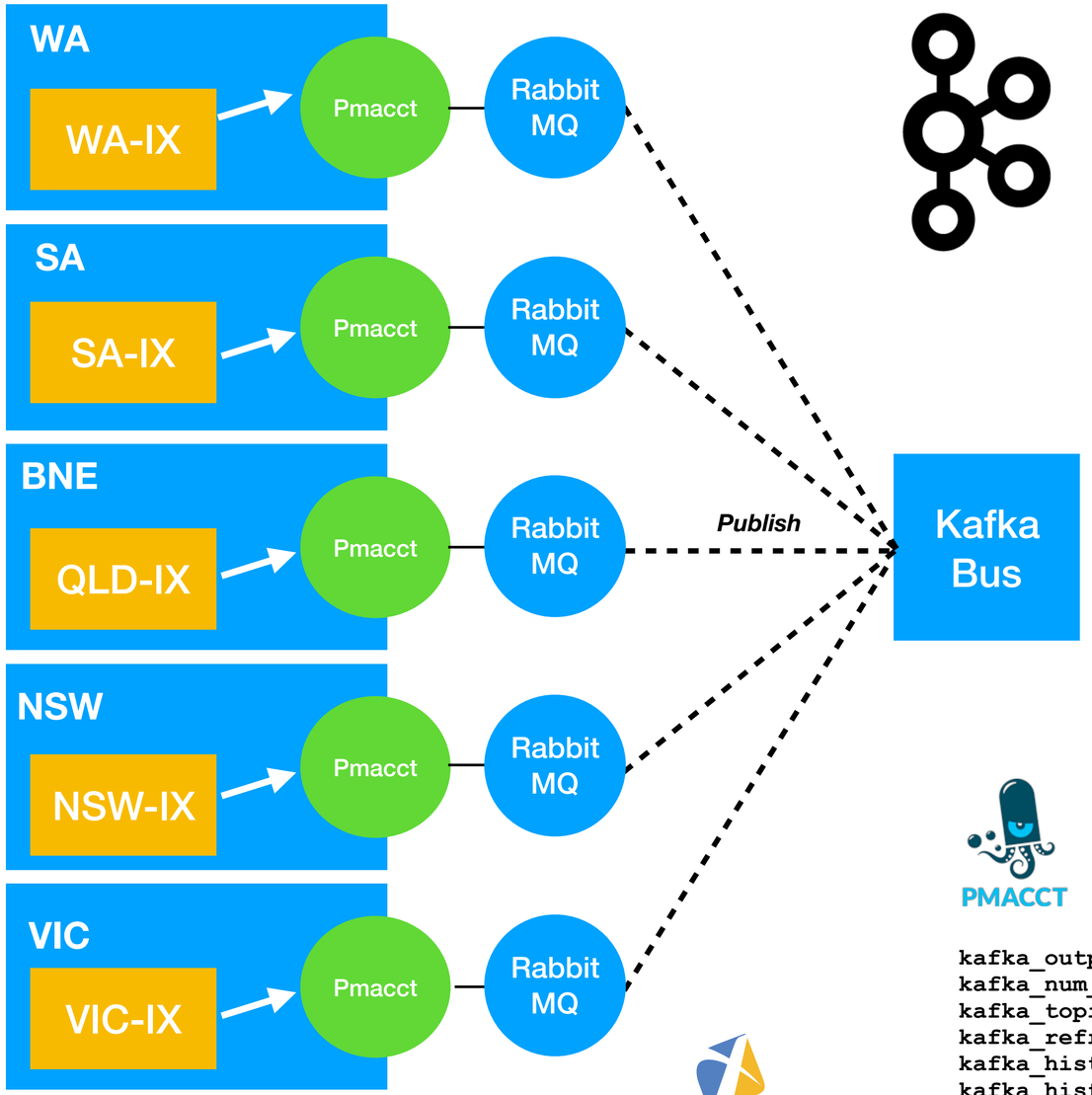
```
daemonize: true
pidfile: /var/run/sfacctd.pid
syslog: daemon
sfacctd_port: 6343
sfacctd_renormalize: true
timestamps_secs: true
timestamps_since_epoch: true

plugins: memory, kafka

aggregate: src_mac, dst_mac, vlan, in_iface, peer_src_ip, etype
```

Transport

- Ensure write-exactly-once to central database.
- Resilience from network disruptions.
- **Solution:**
  - Buffer post collection.
  - Queue to the database.



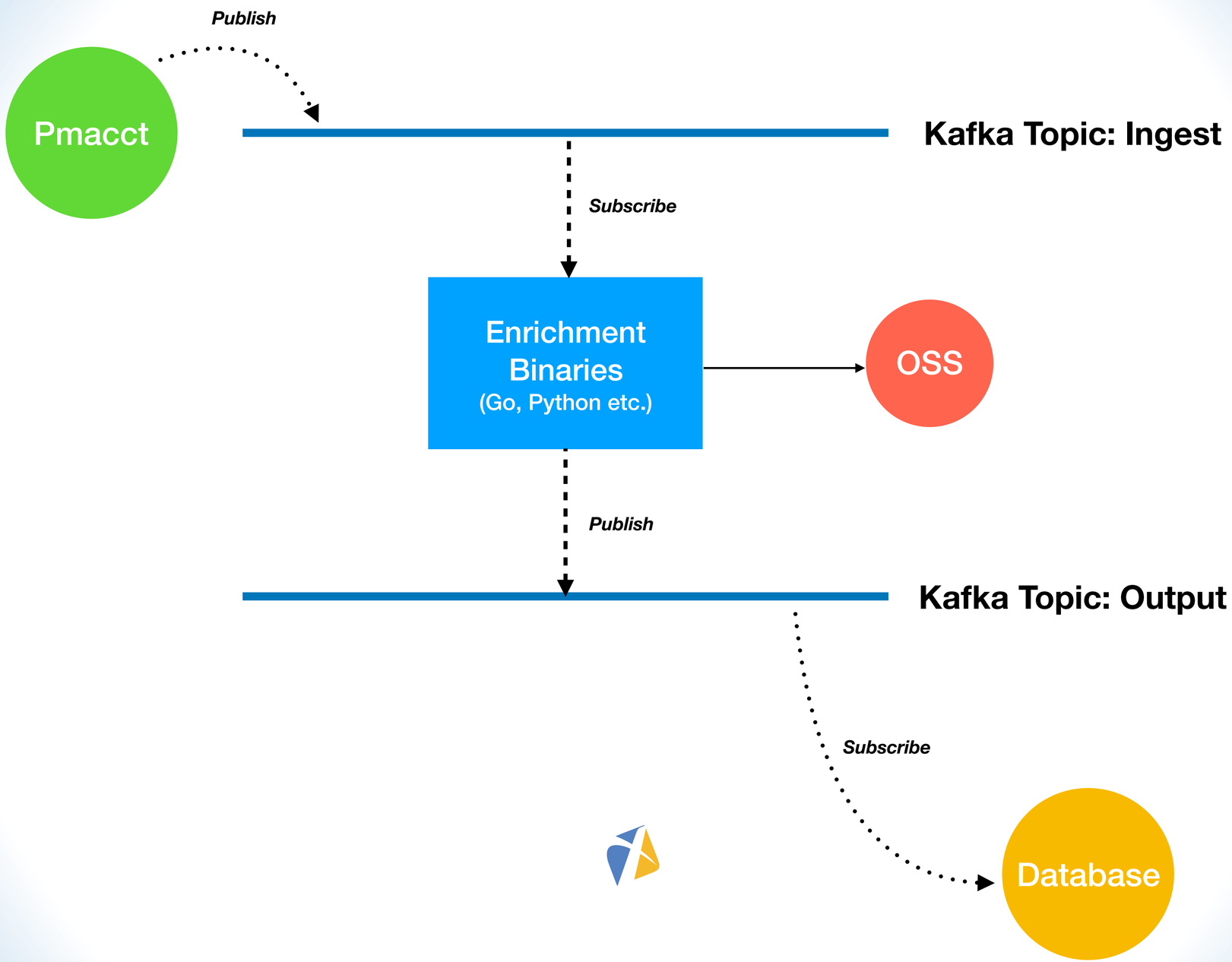
kafka

PMACCT

```
kafka_output: json
kafka_num_protos: true
kafka_topic: sfacct-flow-<state>
kafka_refresh_time: 30
kafka_history: 5m
kafka_history_roundoff: m
kafka_broker_host: 127.0.0.1
kafka_broker_port: 9092
```

Enrichment

- Matching port data to a member
- Additional, future opportunity to add more insight.
- **Solution:**
  - Kafka - moving data between topics in a pipeline.
  - Wide support for tools and frameworks.

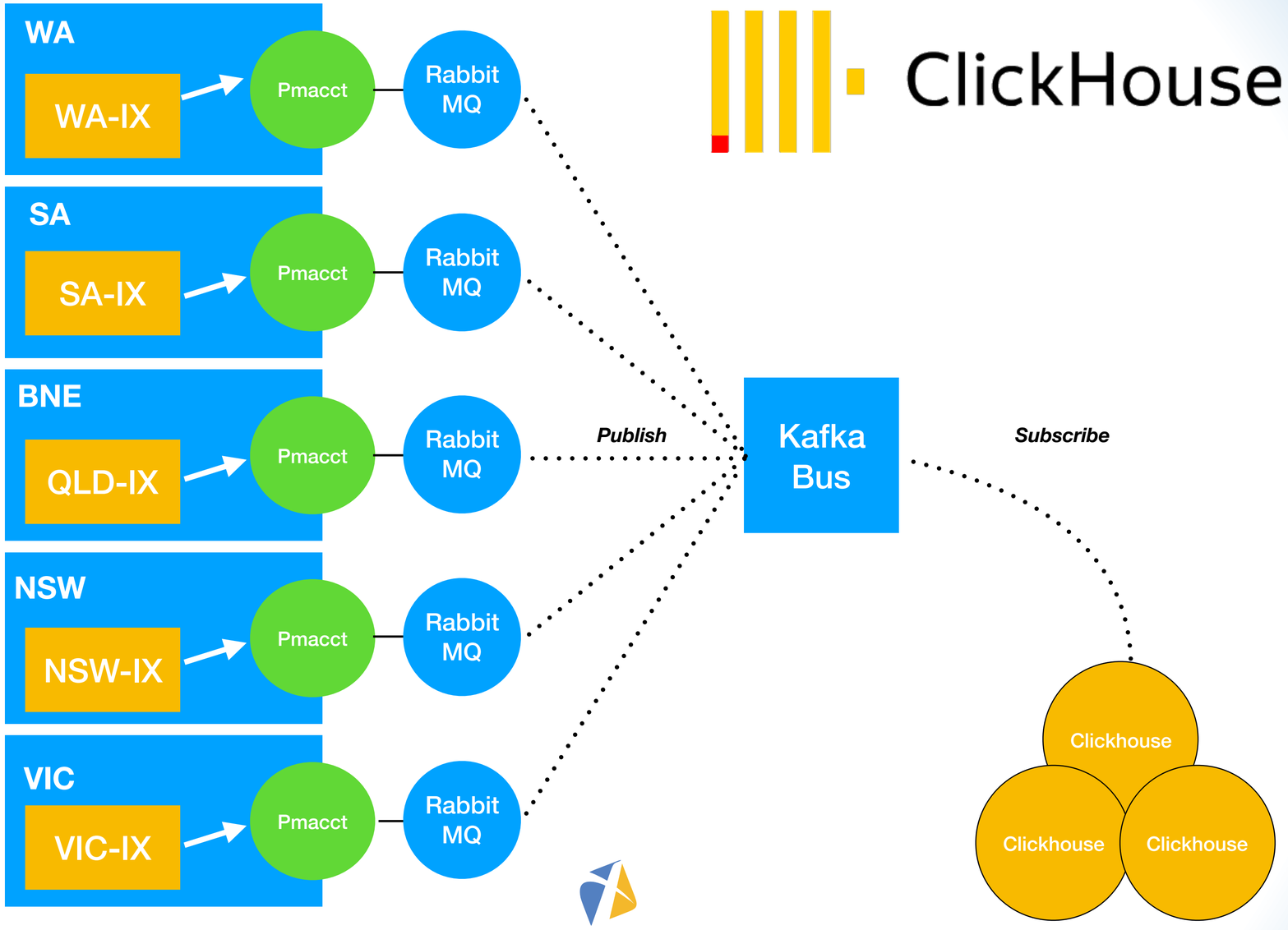

Publish
Pmacct
Kafka Topic: Ingest
Subscribe
Enrichment
Binaries
(Go, Python etc.)
OSS
Publish
Kafka Topic: Output
Subscribe
Database

Storage

- Long-term storage with size-efficient data format.
- Fast query for long-term and short-term data.
- **Solution:**
  - Yanex ClickHouse
  - Columnar database, pluggable engines

Storage

MySQL / Postgres	InfluxDB	Clickhouse
Pros • Well-known • Straight-forward query language	• Timeseries, tag-indexed • Fast queries on tag-sets	• Columnar (not row) based. • Very fast for query on indexes. • Clustering supported in OSS • Can handle time-series
Cons • Doesn't scale easily • Indexing on IPs, MACs etc. is not ideal. • Not focused on time-series.	• OSS version doesn't scale (clustering) • IPs and MACs don't fit tag scaling model.	


WA
WA-IX
Pmacct
Rabbit MQ
SA
SA-IX
Pmacct
Rabbit MQ
BNE
QLD-IX
Pmacct
Rabbit MQ
NSW
NSW-IX
Pmacct
Rabbit MQ
VIC
VIC-IX
Pmacct
Rabbit MQ
Publish
Kafka Bus
Subscribe
Clickhouse
Clickhouse
Clickhouse
ClickHouse

Storage Engine

- Summing Merge Tree
  - Columns designated as “fields” and “value”
  - Every period, all values belonging to a unique tuple of the fields are summed together to produce one set of values.
  - Allows for control over time-period aggregation centrally.

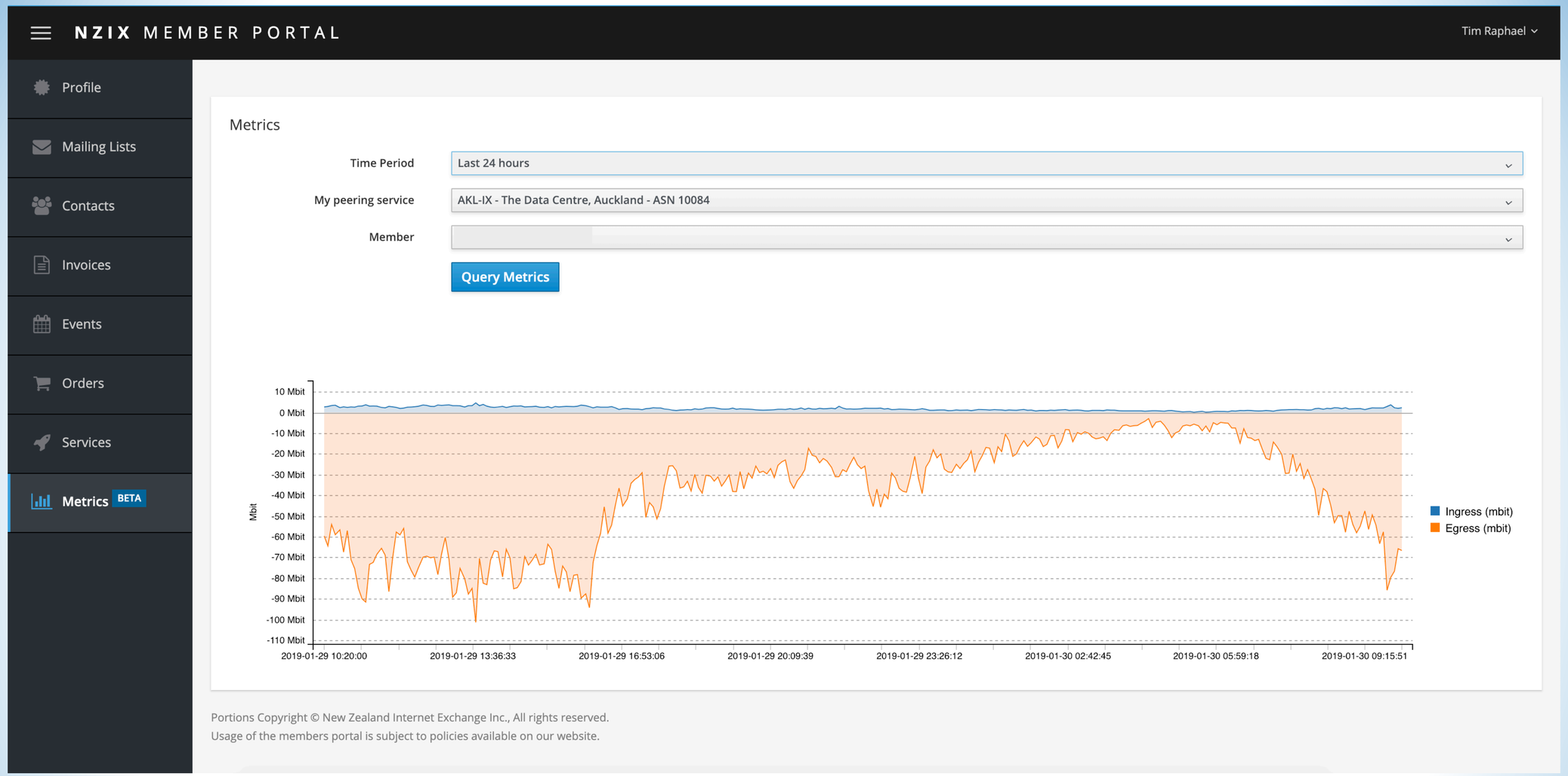
Storage Engine

Analysis

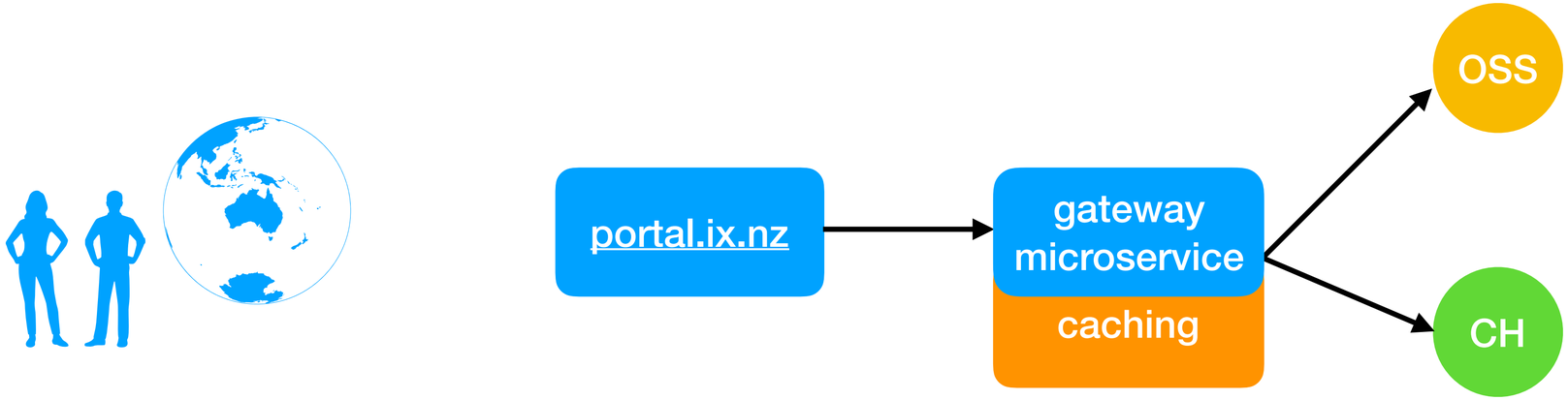
- Support ad-hoc querying
- Custom display from an API.
- **Solution:**
  - Grafana (duh)
  - Clickhouse HTTP API Integration

Demo

NSW Traffic by Destination
All NSW-IX Traffic by Destination MAC
300 Gbps
250 Gbps
200 Gbps
150 Gbps
100 Gbps
50 Gbps
0 bps
20:00
00:00
04:00
08:00
current
04:c5:a4:e3:60:f0 9.09 Gbps
02:04:96:97:a9:5b 8.87 Gbps
74:83:ef:36:e3:89 8.02 Gbps
00:1d:b5:f9:1f:ff 7.73 Gbps
7a:fe:3d:b3:9a:4b 7.22 Gbps
00:04:96:9d:dd:22 7.09 Gbps
44:4c:a8:ce:16:fb 4.23 Gbps
00:04:96:99:44:31 4.23 Gbps
98:5d:82:3c:08:3f 3.98 Gbps
f4:b5:2f:73:40:ac 3.91 Gbps
80:ac:ac:e2:c0:3a 3.68 Gbps
d8:b1:22:96:ef:00 3.67 Gbps
28:a2:4b:f5:f2:a7 3.46 Gbps
00:b8:b3:6e:c3:00 3.34 Gbps
All NSW-IX Data by Source MAC
10 TB
8 TB
6 TB
4 TB
2 TB
0 B
20:00
00:00
04:00
08:00
current
04:c5:a4:e3:60:f0 341 GB
02:04:96:97:a9:5b 333 GB
74:83:ef:36:e3:89 301 GB
00:1d:b5:f9:1f:ff 290 GB
7a:fe:3d:b3:9a:4b 271 GB
00:04:96:9d:dd:22 266 GB
44:4c:a8:ce:16:fb 159 GB
00:04:96:99:44:31 159 GB
98:5d:82:3c:08:3f 149 GB
f4:b5:2f:73:40:ac 147 GB
80:ac:ac:e2:c0:3a 138 GB
d8:b1:22:96:ef:00 138 GB
28:a2:4b:f5:f2:a7 130 GB
00:b8:b3:6e:c3:00 125 GB
NSW-IX Traffic by Source
All NSW-IX Traffic by Destination MAC
300 Gbps
250 Gbps
200 Gbps
150 Gbps
100 Gbps
50 Gbps
0 bps
20:00
00:00
04:00
08:00
current
74:83:ef:36:e3:89 14.87 Gbps
7a:fe:3d:b3:9a:4b 13.96 Gbps
ec:38:73:4b:87:49 12.48 Gbps
f4:b5:2f:a7:1b:c6 12.47 Gbps
00:04:96:a1:05:0e 8.87 Gbps
00:22:83:f0:f7:c0 8.53 Gbps
00:04:96:a1:00:7c 7.09 Gbps
c8:e7:f0:04:eb:c2 6.07 Gbps
54:1e:56:ac:2b:c3 4.68 Gbps
4c:96:14:9f:1f:ca 4.58 Gbps
b0:a8:6e:fd:87:ca 4.58 Gbps
00:12:f2:3f:6c:20 4.50 Gbps
04:c5:a4:e3:60:f0 4.32 Gbps
00:04:96:9b:2e:a0 3.73 Gbps
All NSW-IX Data by Source MAC
10 TB
8 TB
6 TB
4 TB
2 TB
0 B
20:00
00:00
04:00
08:00
current
74:83:ef:36:e3:89 557 GB
7a:fe:3d:b3:9a:4b 524 GB
ec:38:73:4b:87:49 468 GB
f4:b5:2f:a7:1b:c6 468 GB
00:04:96:a1:05:0e 333 GB
00:22:83:f0:f7:c0 320 GB
00:04:96:a1:00:7c 266 GB
c8:e7:f0:04:eb:c2 228 GB
54:1e:56:ac:2b:c3 175 GB
4c:96:14:9f:1f:ca 172 GB
b0:a8:6e:fd:87:ca 172 GB
00:12:f2:3f:6c:20 169 GB
04:c5:a4:e3:60:f0 162 GB
00:04:96:9b:2e:a0 140 GB


NZIX MEMBER PORTAL
Tim Raphael
Profile
Mailing Lists
Contacts
Invoices
Events
Orders
Services
Metrics BETA
Metrics
Time Period
Last 24 hours
My peering service
AKL-IX - The Data Centre, Auckland - ASN 10084
Member
Query Metrics
10 Mbit
0 Mbit
-10 Mbit
-20 Mbit
-30 Mbit
-40 Mbit
-50 Mbit
-60 Mbit
-70 Mbit
-80 Mbit
-90 Mbit
-100 Mbit
-110 Mbit
Mbit
2019-01-29 10:20:00
2019-01-29 13:36:33
2019-01-29 16:53:06
2019-01-29 20:09:39
2019-01-29 23:26:12
2019-01-30 02:42:45
2019-01-30 05:59:18
2019-01-30 09:15:51
Ingress (mbit)
Egress (mbit)
Portions Copyright © New Zealand Internet Exchange Inc., All rights reserved.
Usage of the members portal is subject to policies available on our website.

Portal Consumption

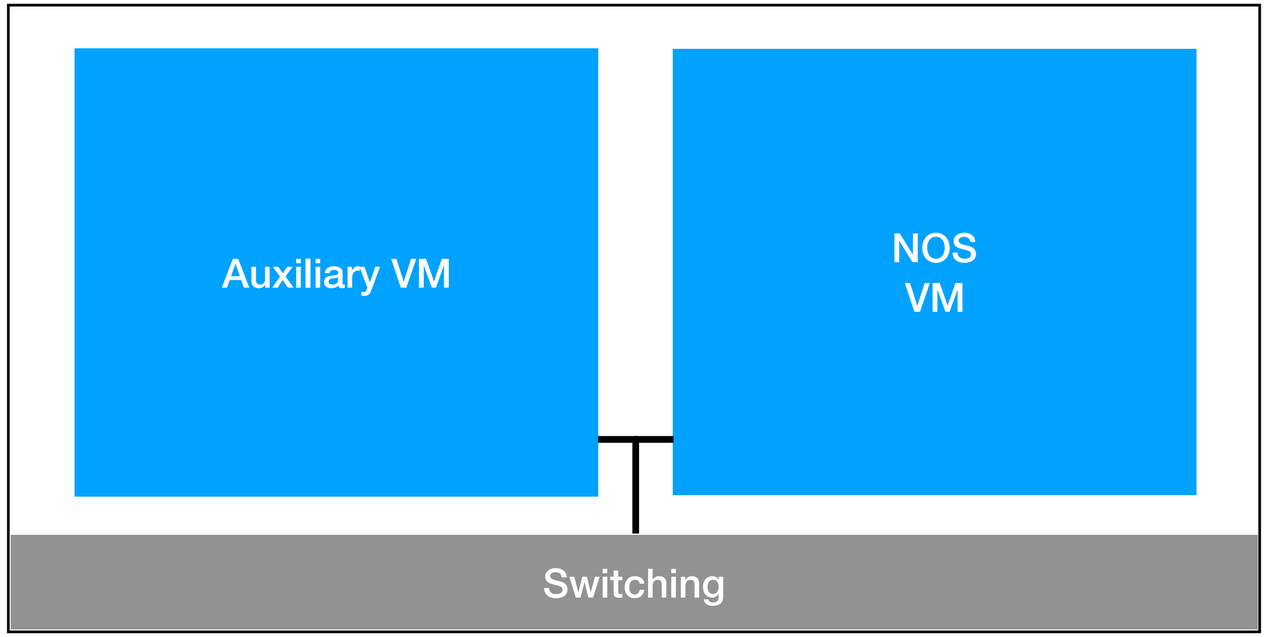


Scaling up the Pipeline

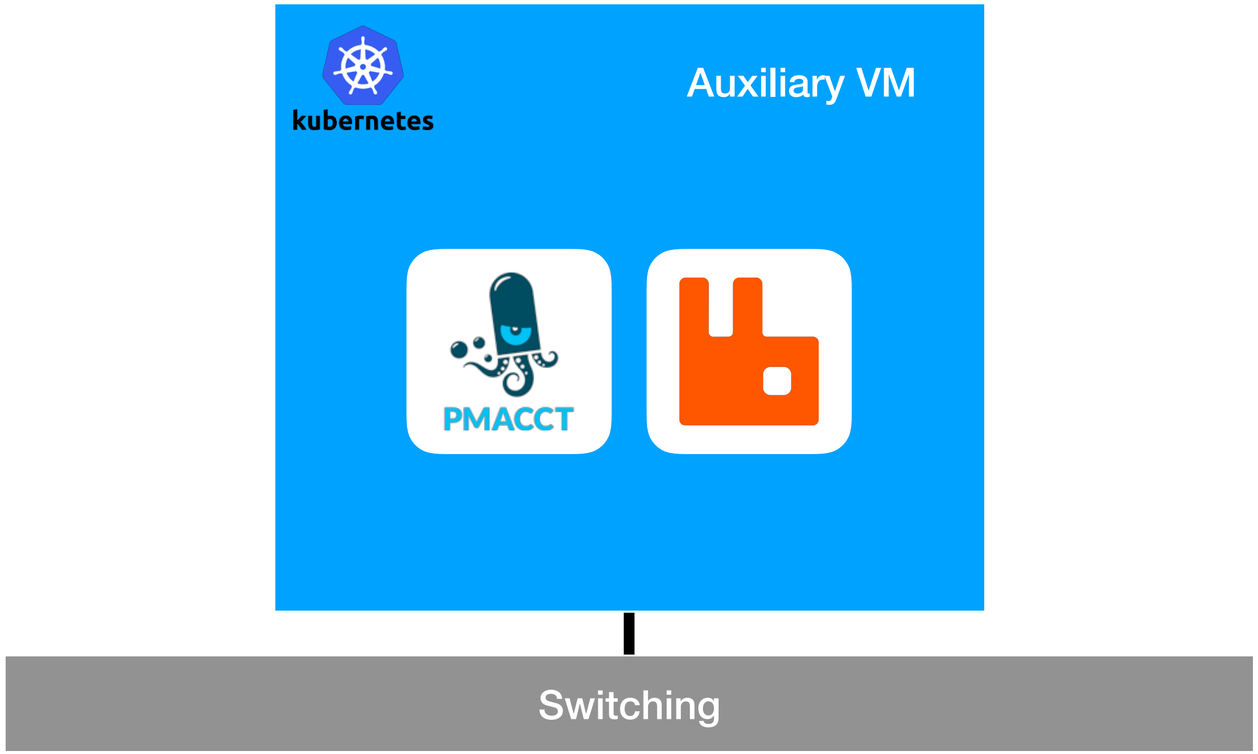
- More flow producing nodes
- Sparse device locations
- Need to process traffic mirrors vs flow data
- Familiarity with micro-service / container architectures

Scaling Up

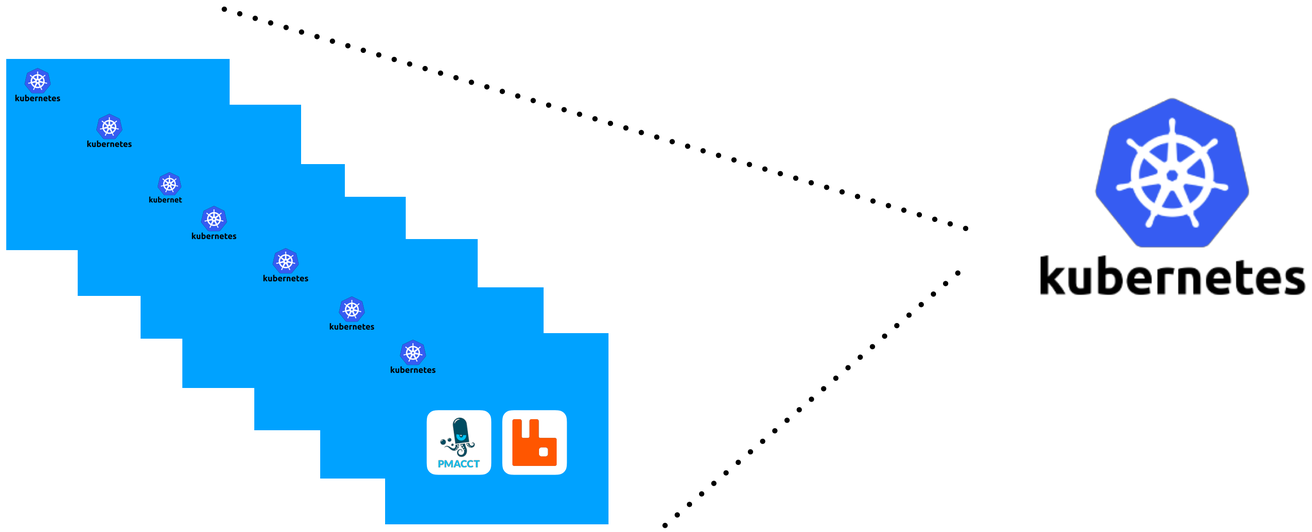
“Whitebox” Switch



Scaling Up



Scaling Up


kubernetes
kubernetes
kubernet
kubernetes
kubernetes
kubernetes
kubernetes
PMACCT
kubernetes

IX | AUSTRALIA

INTERNET PEERING FOR AUSTRALIA